SS-S-291d
July 22, 1968
SUPERSEDING
SS-S-291c
21 December 1956

FEDERAL SPECIFICATION

SHINGLES: ASBESTOS-CEMENT, (ROOFING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers asbestos-cement roofing shingles.

1.2 Classification.

1.2.1 Types and classes. Asbestos-cement roofing shingles shall be of the following types and classes as specified (see 6.2):

Type I - American
- Class A-1 - Individual, Lightweight.
- Class A-2 - Individual, Heavyweight.
- Class B-1 - Strip, Lightweight.
- Class B-2 - Strip, Heavyweight.

Type II - Dutch or Scotch (Including Ranch Design)
- Class C - Square.
- Class D-1 - Oblong, Lightweight.
- Class D-2 - Oblong, Heavyweight.

Type III - French (Hexagonal)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Steel, Strapping, Flat.
QQ-S-790	- Steel Strapping, Round (Bare and Zinc-coated).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

C222 - Asbestos-Cement Roofing Shingles.
C459 - Sampling and Testing Asbestos-Cement Flat Sheets, Roofing and Siding Shingles and Clapboards.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D. C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The asbestos-cement roofing shingles shall conform to ASTM C222, and as specified herein.

3.2 Color. Unless otherwise specified (see 6.2), the exposed surface of the shingles shall be natural cement gray.

3.3 Finish. The exposed surface of the shingles shall be smooth, grained, coated, or textured as specified (see 6.2).

3.4 Type I, American. Type I shingles shall be rectangular in shape with straight or irregular edges.

3.4.1 Class A, individual. Class A shingles shall be designed to be laid with a minimum 2-inch head lap with no side lap, and shall provide double coverage.

3.4.2 Class B, strip. Class B shingles shall be in the form of a strip designed to provide top lap only with single coverage.

3.5 Type II, Dutch or Scotch (including ranch design). Type II shingles shall have straight or irregular edges and shall be designed to be laid with a lap at the top and on one side of each shingle.

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3.5.1 Class C, square. Class C shingles shall be square.

3.5.2 Class D, oblong. Class D shingles shall be rectangular in shape, with the long dimension horizontal.

3.6 Type III, French. Type III shingles shall be square in shape with at least three corners clipped to give the desired pattern when shingles are laid with their diagonals perpendicular to the eave of the roof, with their apex sides lapped.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection of the shingles (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.1.2 For tests. Sampling for tests shall be in accordance with ASTM C459.

4.3.2 Examination. Samples selected in accordance with 4.3.1.1 shall be examined as specified in 4.4.1. AQL shall be 1.0 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1.2 shall be tested in accordance with ASTM C459. Failure of any test shall be cause for rejection of the lot.

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4.4 Inspection procedure.

4.4.1 Examination. The shingles shall be examined for the following defects:

101. Material not in accordance with ASTM C222.
102. Dimensions not in accordance with ASTM C222.
103. Finish not as specified.
104. Color not as specified.
105. Holes for nails and fasteners not in accordance with ASTM C222.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall constitute one defect.
107. Bundles not as specified.
108. Packing not as specified.
109. Strapping not zinc-coated for level A.
110. Marking illegible, incomplete or incorrect for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The shingles shall be packaged in bundles in conformance with table I, secured with two round-wire, 18-gauge steel straps conforming to QQ-S-790, finish 2, for all levels (see 6.3).

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Table I. Packaging

Type	Class	Shingles Per Square	Bundles Per Square	Shingles Per Bundle	Approx. Weight Pounds Per Square
I	A-1	200	10	20	350
	A-2	260	13	20	585
	B-1	88	4	22	290
	B-2	80	5	16	325
II	C	92	4	23	252-265
	D-1	80	4	20	255
	D-2	80	5	16	255
III	-	88	4	22	240

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. The shingles, packaged as specified in 5.1 shall be packed as specified in table II. in a wood-cleated plywood box conforming to PPP-B-601, overseas type, style A or B, type 3 load. The weight limitations shall be waived on type I, class A and type II, class A and B boxes. Strapping shall be zinc coated conforming to QQ-S-781.

Table II. Packing and Palletizing

		Level A	Level B			
Type	Class	Squares	Squares	Courses	Bundles per Course*	Approx. Weight
I	A-1	2	8	10	8	2800
	A-2	2	8	8	15	4680
	B-1	3	8	8	4	2320
	B-2	3	8	10	4	2600
II	C	4	15	12	5	3975
	D-1	4	12	12	4	3000
	D-2	4	12	12	5	3060
III	-	4	15	12	5	3600

* Top course may be less than quantity indicated per course to load only full squares per pallet.

5.2.2 Level B. The shingles, packaged as specified in 5.1 shall be palletized in quantities specified in table II. Palletization shall be in accordance with MIL-STD-147, Palletized Unit Loads, except that horizontal straps shall not be used.

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5.2.3 Level C. The shingles, packaged as specified in 5.1 shall be shipped in a manner to assure carrier acceptance and safe delivery to destination at lowest rates, in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. Bundles and containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The asbestos-cement shingles are designed to provide the weather-exposed surface on roofs of buildings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Types and classes required (see 1.2.1).
(c) Color required when other than natural cement gray (see 3.2).
(d) Finish required (see 3.3).
(e) Level of packing required (see 5.2).

6.3 Packaging. A single method of packaging, i.e. bundling, is applicable to all levels. For purposes of the preservation and packaging level marking, this may be A, B, or C, as applicable (see 5.1).

Custodians:

Army - ME
Navy - YD
Air Force - 84

Preparing activity:

Army - ME

Review activities:

Navy - YD
Air Force - 85

User activity:

Army - CE

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